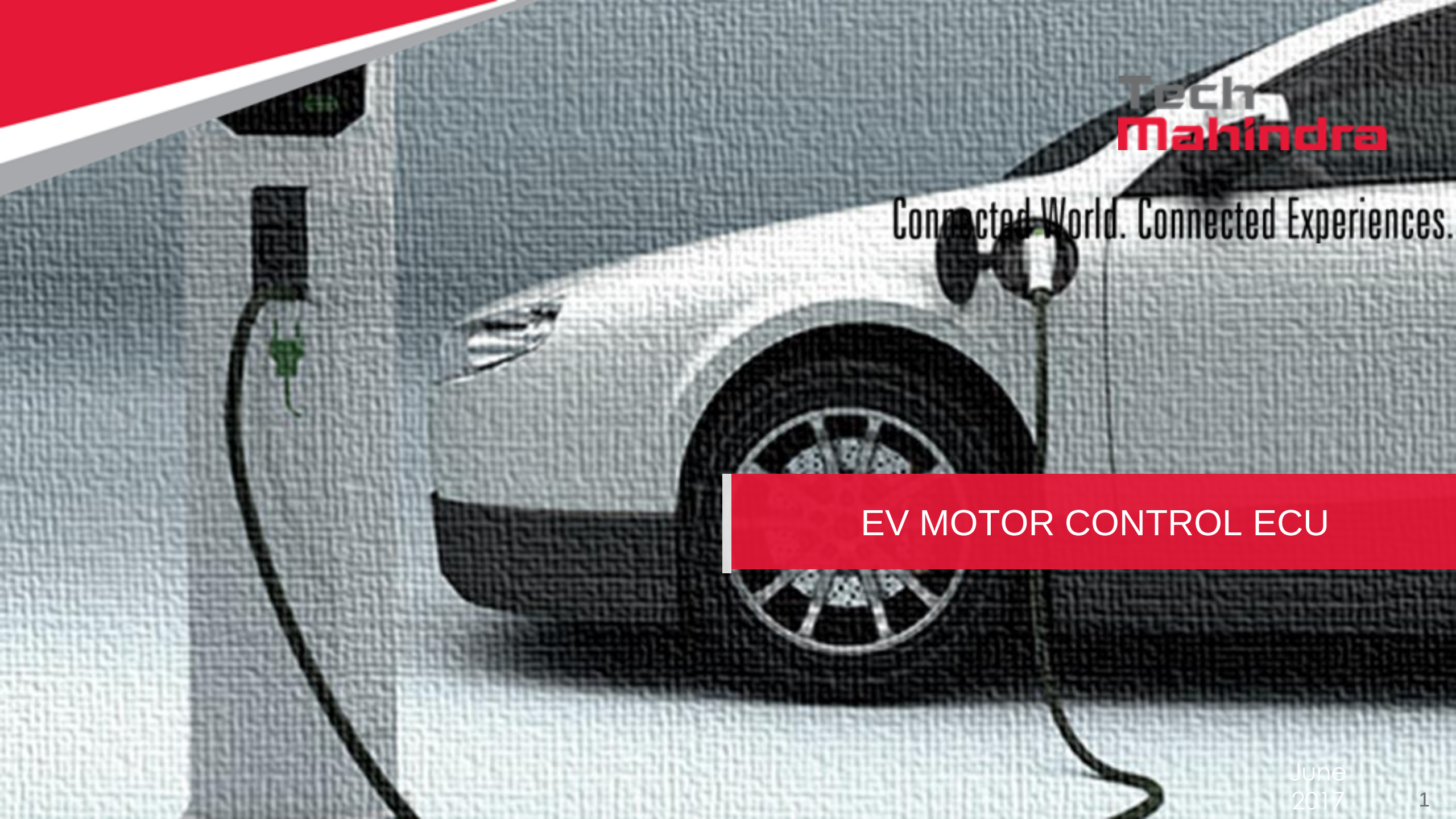




**Tech
Mahindra**

Connected World. Connected Experiences.

EV MOTOR CONTROL ECU

OUR OFFERINGS IN EV

Hybrid/ Full Electrification



Full Electrification & Conversion of Gas to Electric using Ready to Use Powertrain Kits

In-Vehicle

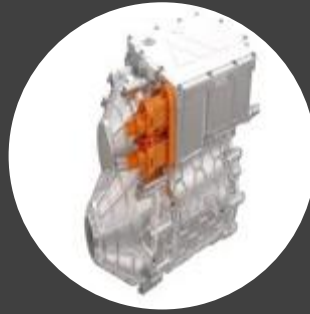
Build/Benchmark your Kit



Create the Powertrain Kit based on Customer Specifications using Ready to Use Kits and Benchmarked Components

In-Vehicle

Components Development



Design Development and Validation of Components (both Software and Hardware)

In-Vehicle

Validation Services



Component & Vehicle Validation/Certification/Homologation

In-Vehicle

Energy Systems



Custom Chargers, Energy Solutions

Out-Vehicle

Connectivity and Technology Bundles - Connected Car, Mobility Services, Analytics & More ..

MILD HYBRID MOTOR CONTROLLER ECU – CASE REFERENCE

Industry Context

The proliferation of the Hybrid & Electric Vehicle is dependent on the performance of an electric motor and power electronics components

Business Objective

- Advanced motor controller algorithms to meet vehicle level requirement
- Highly efficient inverters / converters to generate quality power supplies
- Powerful hardware to support sophisticated algorithms
- Model based software development life cycle

Solution / Approach

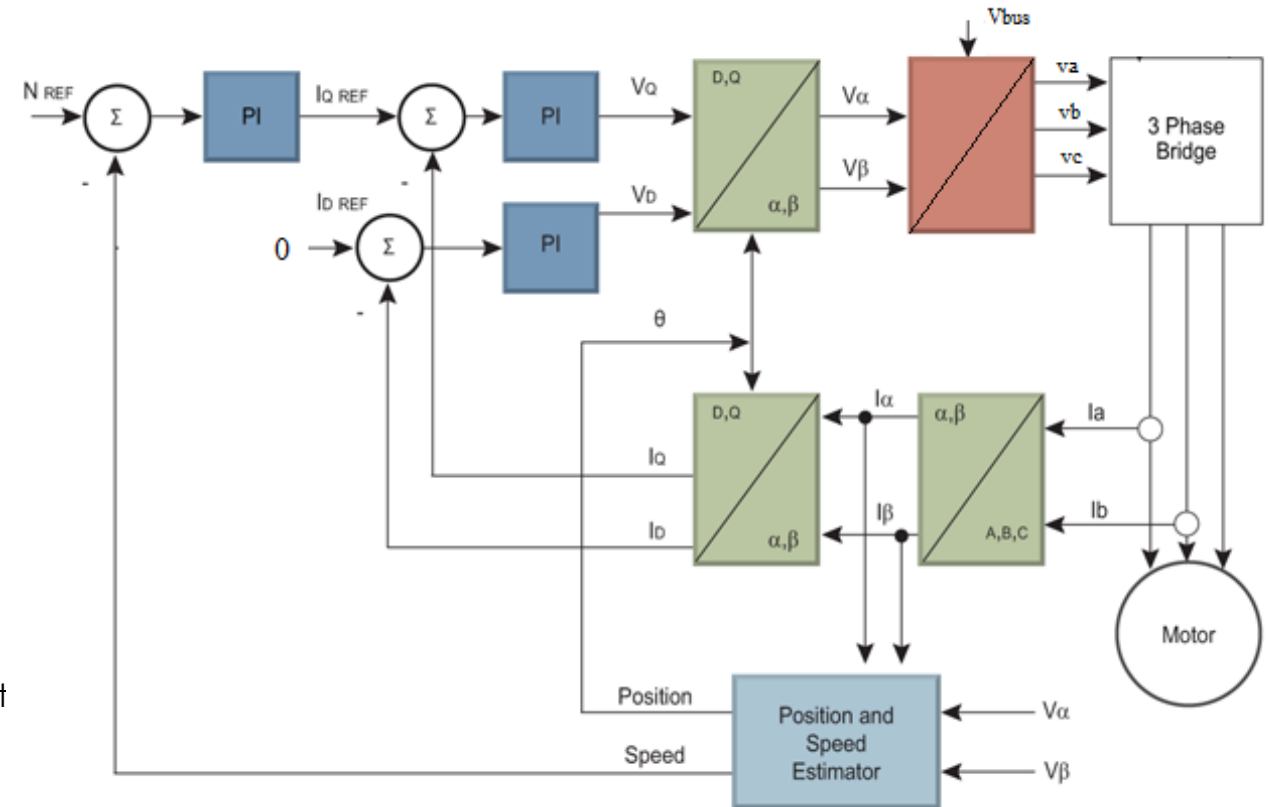
- High Performance Torque Control Scheme
- High Torque & speed control capabilities
- Highly efficient power converters
- Regenerative braking capabilities
- ISO 26262 compliance
- AUTOSAR compliance
- EMI / EMC Compliance

Business Impact

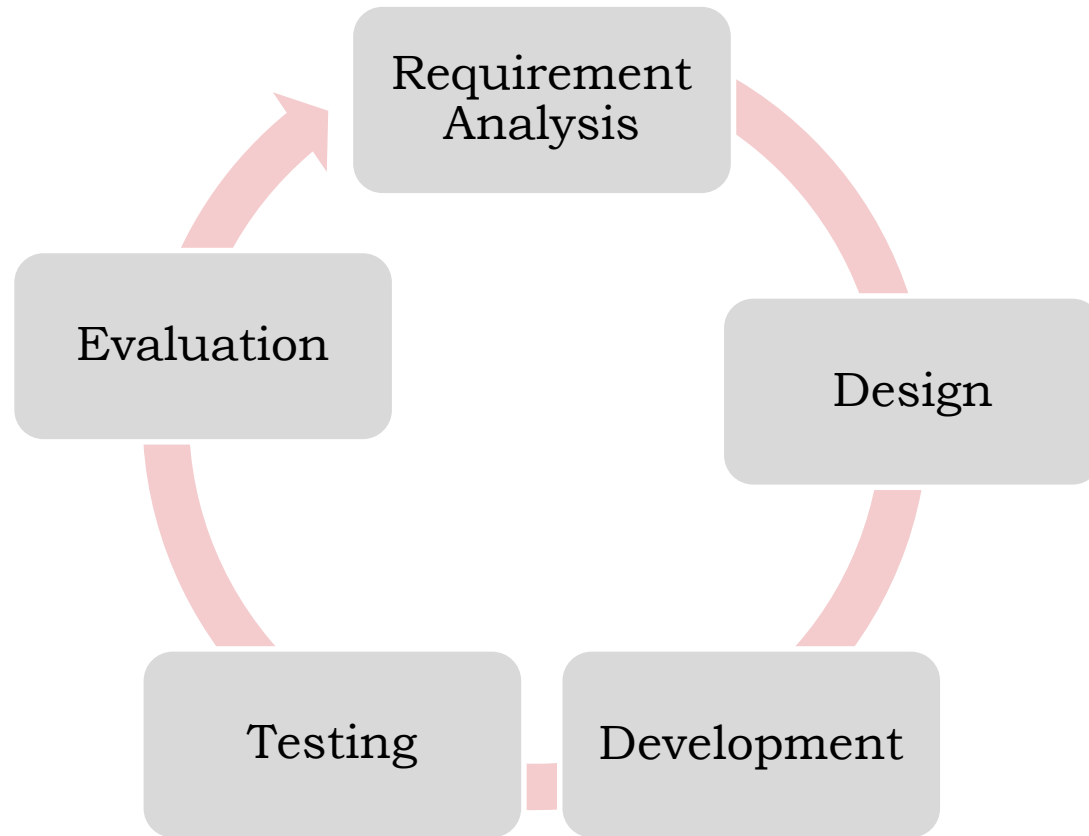
- Successfully developed and tested for different vehicle operating conditions
- Scalable Architecture
- Advanced control algorithms
- Design and development of highly efficient power converters

Value Addition

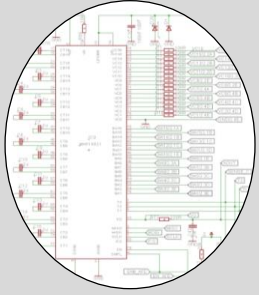
- SVPWM, Anti-Windup Scheme and Field Weakening Control



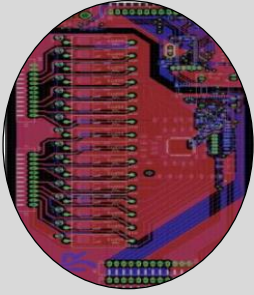
SOFTWARE DEVELOPMENT WORKFLOW



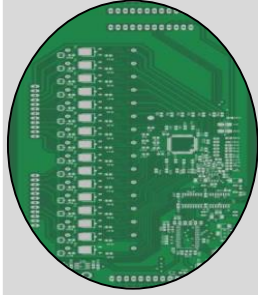
HARDWARE DEVELOPMENT WORKFLOW



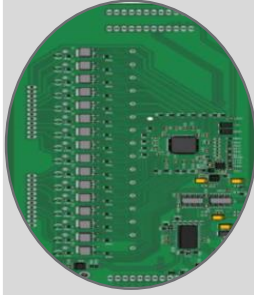
Define
Architecture



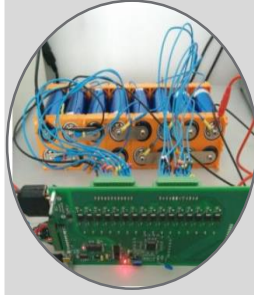
Select Key
Components



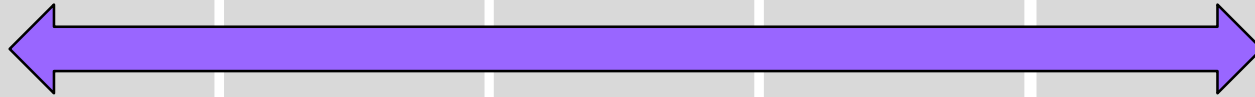
Schematic
Design



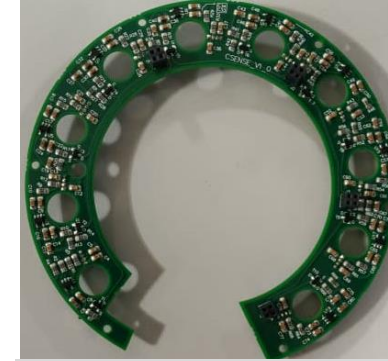
PCB Layout



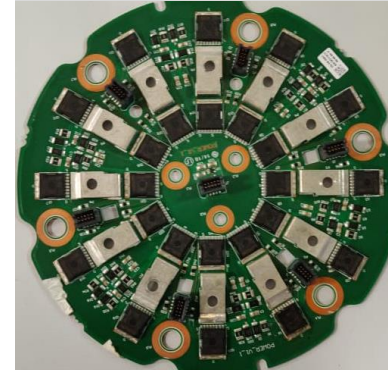
Assemble, Test
& Debug
EMI/EMC
Conformance
Test



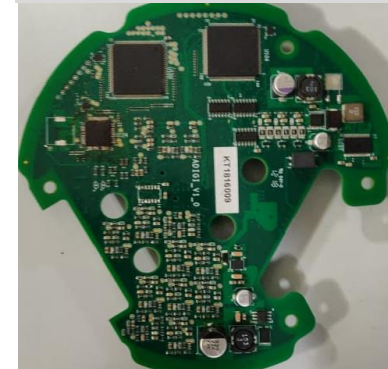
Hardware Design Tools:
Mentor Graphics, OrCAD, Simetrix, pSpice



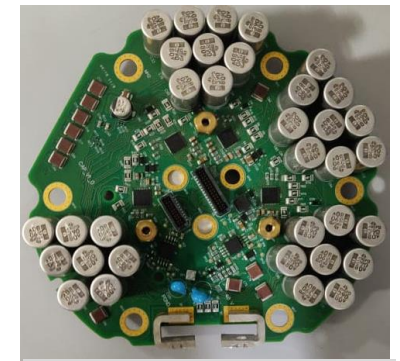
Current Sensor



Power



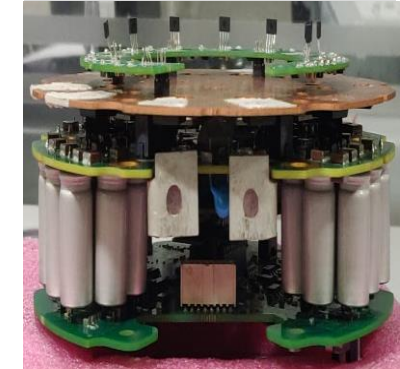
Main Controller



Capacitor

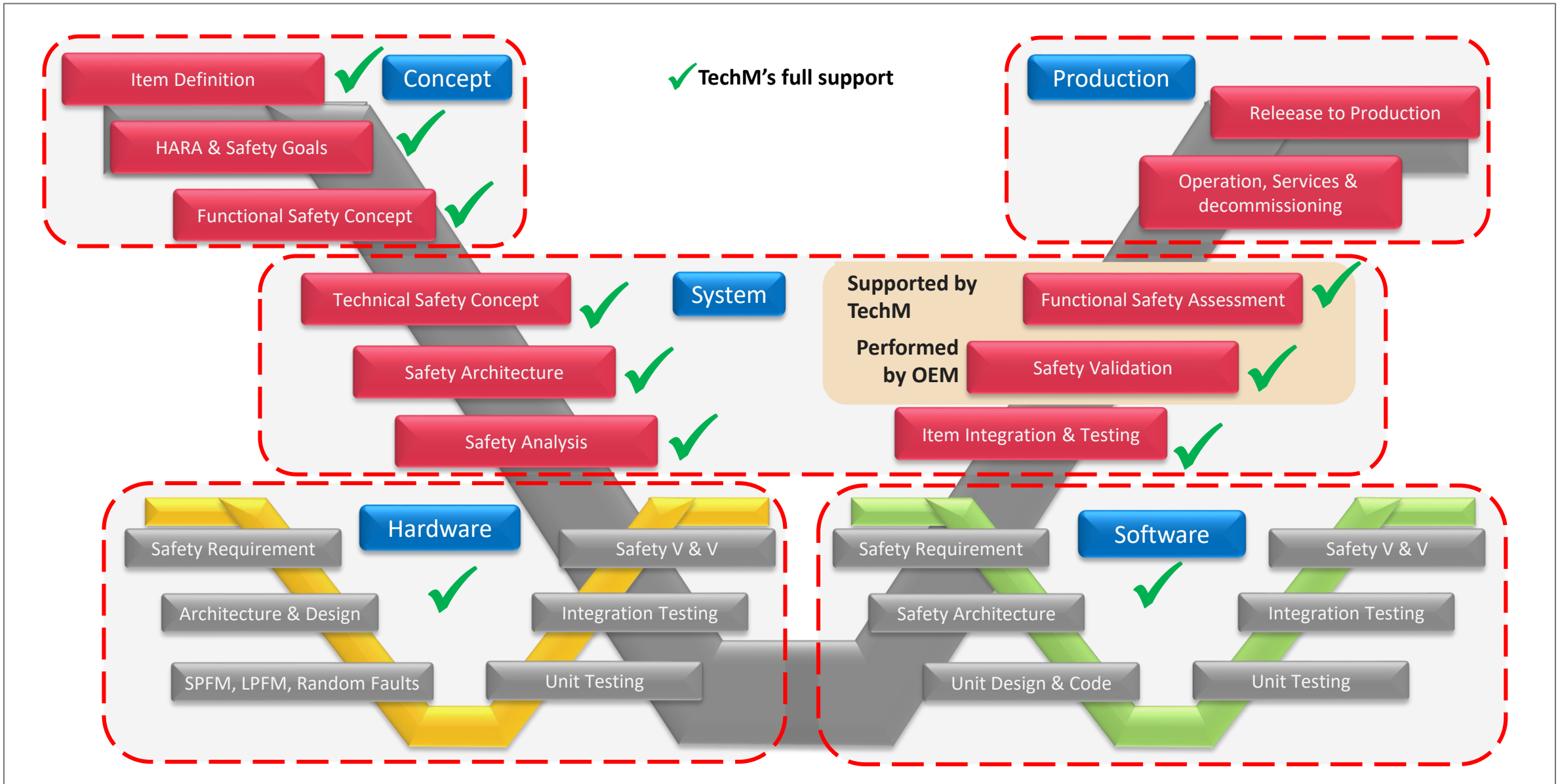


Isolation



Motor Controller ECU

FUNCTIONAL SAFETY – ISO 26262



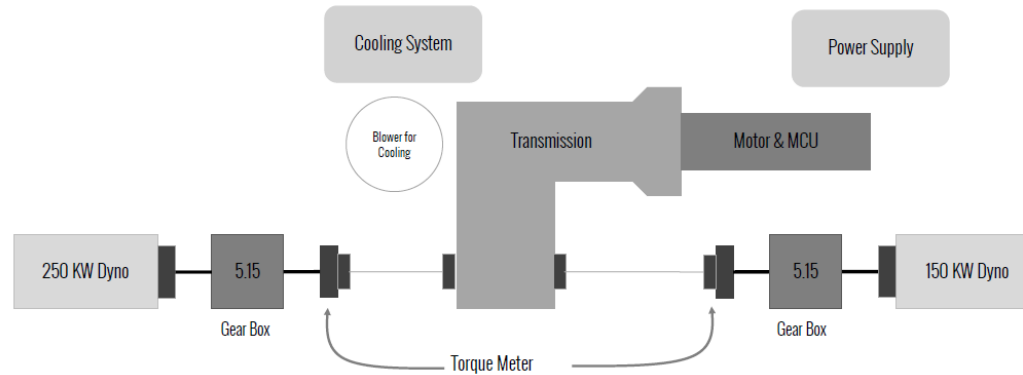
VALIDATION & TESTING OF ECU AND MOTOR



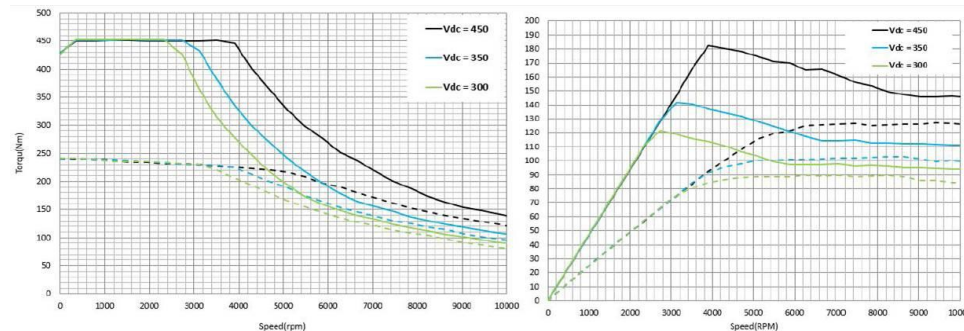
80kW, 152 Nm



150kW, 250 Nm



- Variable Torque Testing
- Back to Back Dyno 250kW
- Up to 3000 NM Torque
- Cost Effective
- Our own IP



Speed Vs Torque

Need of Dyno Testing

Determining performance of Electric Motor

Input Power

- Voltage up to 650V
- Current up to 600A

Speed

- Ramp up time
- Running Speed
- Allowable Speed Variation
- Speed Vs Efficiency

Load Profile

- Starting Torque
- Peak Torque
- Continuous Torque
- Duty Cycle
- Speed Vs Torque
- Regenerative Braking

Environment

- Temperature
- Ingress Protection

Data Analysis

- 4 Channel Power Analyzer
- LabVIEW Software



Thank You.

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